Inductive Reasoning as Art, Not Algorithm:
Criteria and a Dated Genealogy for Inference-Driven Discovery

Ajax Benander*

First drafted July 2026
This version: October 2026 — draft in preparation

Abstract

Philosophy of science has studied discovery mostly through third-party reconstructions from the notebooks of the dead; prospective, self-deposited, real-time records with public certificates have been scarce, because the discipline's founding hygiene — the separation of the context of discovery from the context of justification — hardened into a publication genre that discards such data at the source. This essay improves the evidential situation from an unusual position: the author possesses the data. Two things are put on the table. First, a set of named criteria and theses — the Strong Implication Criterion and the Strong Rooting Criterion on ontological adequacy; the substance/reasoning axis separating failed from successful theories; the distinction between ontological and methodological revolutions; the two-stage, charming-then-structural shape of inference-driven breakthroughs; the shades-of-similarity account of why deep unknowns hide, deepened to the lexicon by the phenomenon here named linguistic masking; and the thesis that inductive reasoning is an art form of which deductive reasoning is part of the artwork — each stated in a displayed box and argued first on external historical cases (Newton and Maxwell, Einstein's road to general relativity, Feynman's nanotechnology lecture, string theory, superdeterminism) before touching the author's own work. Second, a dataset: the dated genealogy of one framework's construction, from a preschool epiphany through journal entries of 2022–2023 excerpted verbatim to four public prints (January and May 2023, 2024, 2026) whose word-frequencies are recomputable by any reader. The record is graded rather than haloed: third-party-dated where the anchors are public — an archive's own version trail, DKIM-signed correspondence — author-hosted where the prints are self-deposited, their day-and-second precision resting on file metadata under the author's control, and memoir in grade where the exhibits are childhood recollection. The theses are then run live on the dataset, and a companion paper audits the same record claim by claim. The position taken on the old firewall is split and exact: dropped for content, because discovery context is evidence; re-erected for normativity, because there are no rules of discovery, while the rules of checking — exhibited here in an AI-era provenance apparatus — should be exacting.

1 The claim, and a disclosure

The discovery side of science is not unstudied — the notebooks of the dead have been read closely and well: Gruber on Darwin [3], Tweney on Faraday [4], Holmes on Krebs [5], Nersessian on

*Email: akb9408@rit.edu, Website: `https://aethus.org`. This paper collects and develops, as a standalone essay in the philosophy of science, methodological material distributed through the long companion manuscript [1], whose dated genealogy it draws on as data. Unlike the framework extractions, this paper is deliberately first-person: the documented genealogy is the dataset, and the voice is part of the method. The record is audited, with a coding protocol and claim-by-claim tests of later accounts, in a companion paper [2].

Maxwell [6], the discovery-logic debates collected by Nickles [7], and Hanson before all of them [8]. What the field has scarcely possessed is data of a different kind: prospective, self-deposited, real-time records kept by the discoverer while the discovering happened, with public certificates a reader can recompute.[1] The shortage of *that* is self-inflicted. The founding hygiene — Reichenbach's separation of the context of discovery from the context of justification [13], with Popper's dictum that there is no logical method of having new ideas as its normative twin [14] — was adopted so thoroughly that it hardened into a publication genre: finished papers present finished structures, the scaffolding is struck before any reader is allowed inside, and the records of how a framework actually came to be are treated as private exhaust rather than as evidence. Theories of scientific production thus get built from completed artifacts and third-party reconstruction, and features of the artwork are projected onto the making. The present dataset's delta over the notebook studies is access — real-time rather than reconstructed, self-annotated rather than inferred, certified rather than curated — and its cost, conceded up front, is worse selection bias: one subject, self-selected, self-reported. I will argue below that this projection is the central category error in the theory of inductive reasoning; but before arguing it, I want to name the evidential situation it has produced, because this paper is constructed specifically to improve that situation from an unusual position: I happen to possess the data.

This paper puts two things on the table. The first is a set of named criteria and theses — the Strong Implication Criterion and the Strong Rooting Criterion (§2); the substance/reasoning axis on which failed and successful theories separate (§3); the distinction between ontological and methodological revolutions (§4); the two-stage, charming-then-structural shape of inference-driven breakthroughs (§5); the shades-of-similarity account of why the deepest unknowns hide (§6); and the thesis that inductive reasoning is an art form of which deductive reasoning is part of the artwork (§8). Each is stated in a displayed box so that it can be cited, challenged, or used without archaeology, and each is argued first on external historical material — Newton and Maxwell, Einstein's road to general relativity, Feynman's nanotechnology lecture, string theory, superdeterminism — before it is permitted anywhere near my own work. The historical tellings those cases rest on are themselves contested, and nothing here pretends otherwise: the readings used are the standard ones, chosen for legibility, and where a rival historiography reads an episode differently — Newton's synthesis as more collective and more gradual than the telling here, for one — the thesis is owed the harder reading rather than the convenient one.[2] The second is a dataset: the dated genealogy of one framework's construction, mine, running from a Google Doc written at 7:55 PM on July 24, 2018, through real-time journal entries of August 2022 and February 2023 reproduced verbatim in §9, to four public prints (January and May 2023, 2024, 2026) whose contents — down to word frequencies — are checkable by any reader against deposited text. The theses are then run live on the dataset, and the reader is invited to audit the fit rather than take my word for it.

Now the disclosure, because the paper's structure makes it mandatory rather than polite.

[1] Two bodies of work come nearest, and each locates the scarcity by what it lacks. Dunbar recorded scientific reasoning as it happened, an observer attending the weekly meetings of molecular-biology and immunology laboratories for a year [9, 10]: real-time, but recorded by a third party and confined to the group stage. The Polymath projects put mathematical discovery in progress online in the participants' own timestamped comments, dead ends included [11, 12]: real-time and self-deposited, but public from the first move, on problems posed in public, the first of them settled in weeks. Neither holds one discoverer's private stages across years, which is what the present record adds, its private leg dated by its author (§9.1).

[2] The standard to aim at is the one §7.1 sets against itself: cases coded, coding weaknesses named, corrections forced against material already installed. The classic-case treatments are vignettes by comparison — illustrations calibrated against standard tellings, not independent historical results — and a reader who rejects a telling should read the corresponding thesis as owing a better case, not as refuted by the disagreement.

Every thesis defended here classifies my own framework favorably: on my own axis my framework falls on the reasoning side; in my own taxonomy of revolutions it is of the ontological kind; my own genealogy exhibits the two-stage shape my own model predicts. That is an occupational hazard of doing philosophy of science from inside a research program, and I would rather name it than perform innocence of it. The mitigations are structural. The external cases come first, and the theses must earn their keep there before touching anything of mine. The dataset is dated, deposited, and checkable — including its second column, the things the early records got wrong or dressed differently, which I have kept on the page rather than pruned. And the theses are stated sharply enough to fail: a reader who finds a substance-adding theory that succeeded, a deep unknown that arrived wearing no camouflage, or an axiomatization of inductive practice that did not smuggle its era's assumptions has found a counterexample, and I have written the boxes so that such a reader knows exactly what they have hit.

There is a second disclosure inside the first, and it changes the logic of the hazard rather than merely its optics: for the two criteria of §2 — and, per the genealogy's own earliest exhibit, for the axis in embryo — the direction of dependence runs from the methodological intuitions to the framework, not the reverse. The intuitions came first — the dissatisfaction with superdeterminism, the rooting instinct that the constraint must live one layer up — and the framework was deliberately constructed to land them: built to the spec, not scored by a curve fitted afterward, which is what §2's compass sentence already asserts in passing and is here asserted in full. That dissolves the post-hoc charge at a price this paper accepts explicitly: a construction engineered to pass a test contributes nothing to the test's validity, so my framework's favorable scores are zero evidence for the criteria, which must earn their entire keep on the external cases — the external-cases-first structure above is that logic enforced, not a courtesy. What the design direction buys is the converse reading: the scores report that the construction met its target, and they carry exactly as much weight as the criteria independently earn. The priority claim is itself dateable and graded like everything else here — certified where the intuitions' early statements survive in dated entries, memoir-grade where they do not. And one thesis stands exempt from the charge of design by its nature: the two-stage shape of my own genealogy cannot be a design artifact, because being surprised by one's own future self is not a specification whose satisfaction anyone can engineer. It is not exempt from retrospective organization, since the episodes in which the surprise is found are selected and classified afterward, by me (§5).

One completion of this disclosure was, in an earlier draft, left for the reader, and my own thesis about predictable readings says that is exactly backwards, so here it is owned. §4 argues that the ontological move has historically required a single mind; the same section finds quantum theory's ontological move unratified; and the companion papers claim to perform precisely the unscrambling that move names. Any attentive reader will complete the syllogism, and I will not perform not having completed it myself. Its status is the entire point of stating it: the conclusion is not a finding of this paper — every thesis here stands or falls identically whether the companions succeed or fail — but a claim those papers must earn under independent refereeing, where it is at present unearned; and this paper's contribution to that adjudication is the dataset alone, which will show, whichever way the adjudication goes, what the attempt looked like from inside.

One more declaration, of genre. Unlike the framework extractions that share this corpus, this essay is deliberately first-person, and the choice is methodological rather than stylistic. The memoir-as-data method has a small but venerable lineage — Einstein's autobiographical notes, Poincaré's essays on invention [15], Hadamard's study of mathematical creation [16], Polanyi's personal knowledge [17] — and what that lineage understood is that the first-person record is not a rhetorical liability to be laundered out but the only place certain evidence exists at all. My position

on the old firewall is therefore split, and I want to state it exactly: I drop the discovery/justification separation for *content*, because the context of discovery is evidence and discarding evidence is not rigor; and I re-erect it for *normativity*, because there are no rules of discovery worth the name, while the rules of checking should be exacting. The paper practices both halves: the genealogy below is as unguarded as the original notebooks, and every claim that can be checked is routed to something datable, deposited, or recomputable.[3]

1.1 The shape of the defense

A paper advancing this many named claims meets a calibrated prior: most papers of that shape are confused, and a referee is right to hold the prior until the page rebuts it. The rebuttal offered here is not the count but the closure: the claims constrain one another concretely, and the interlocks are on the page, citable by section, rather than asserted in a preface. The rooting criterion derives the implication criterion outright — a toy derivation, marked as such, but a derivation and not an analogy (§2); the art thesis polices even this paper's newest guidance into negative form, forbidding the sign-flip that would have axiomatized the proximity corollary (§7, §8); the corollary's vacancy pricing is cashed on the implication criterion's dated first-contact yield (§2); the one-way conditional of §7 governs both that corollary and that yield, each installed with the restriction written into it; the closing prescriptions are licensed by the art thesis's own line that heuristics are not axioms (§12); the firewall's two halves are not argued but practiced — the genealogy unguarded, the certificates exacting (§9.1); and the whole apparatus is applied to the paper itself and terminates at an external gate rather than in further text (§11); and the measurement corollary closing §8 is derived from the art thesis rather than asserted beside it. Two consequences follow, one against me and one for me, and both belong on the page. Against: correlation is shared risk — these members issue from one mind and one framework, so damage propagates, and ten interlocked claims are nearer to one structure with ten load-bearing members than to ten independent results; a referee who wounds a member should expect the wound to travel, and is invited to test how far. For: coherence is audit efficiency — any member can be tested against the others' stated constraints, and a structure whose parts visibly object when one is removed cannot be dismissed claim-by-claim without the objections going on the record. The conditional runs the usual single direction: interlock buys non-arbitrariness and checkability, never correctness; a coherent structure can be coherently wrong, and the external cases and the dated record remain the only places that question is decided. The firewall the paper splits should be stated as the one theorem the whole essay rests on, in the form a reader can check. Let D be discovery-record evidence, H_P a hypothesis about the process by which a theory arose, and H_T the hypothesis that the theory is true. The split firewall says: $P(H_P \mid D)$ may legitimately differ from $P(H_P)$—the record is evidence about the process, about what was written and thought when, about whether idea A preceded idea B, about whether a later reconstruction matches the contemporary notes—while D moves $P(H_T)$ only through what it shows about the process. The principle must not be stated one notch too strong, and the correct strength is this: D is directly evidential about H_P, and D does not confirm H_T *merely because the discovery narrative has a*

[3] Reflexivity is not new to the study of science, and the positioning is owed. The sociology of scientific knowledge made it a principle — the Strong Programme's fourth tenet [18] — and Ashmore's study performed its own thesis on the page [19], a wing whose textual self-reference famously spiraled toward regress, which is part of why the mainstream kept both its distance and its self-exemption. What this paper practices is reflexivity of a different kind: certificate-bearing rather than literary — self-application that terminates in recomputations over frozen deposits and at an independent gate, rather than in further text about itself — so the regress that made reflexivity a cautionary tale finds no purchase here, and the mainstream's reason for abstaining does not transfer.

favored shape. It is not that process evidence can never bear on truth. A record can show that a theory was fitted blind before a held-out dataset was seen, that an apparent derivation was reverse-engineered from its answer, that independent routes converged on one result, or that reporting was selective—and each is a fact about the process that rationally moves confidence in H_T with no newly precommitted prediction, because each bears on the reliability of the method that produced the theory. Literal Bayesian screening, $P(H_T \mid D, E, M) = P(H_T \mid E, M)$, therefore holds only under the assumption $H_T \perp D \mid (E, M)$, with E the object-level evidence and M the independently validated information about the method's reliability; where D is itself informative about M, the assumption fails and the record bears on truth through that channel. What the firewall forbids is the direct channel, from the shape of the story to the truth of its conclusion. Everything the dataset of §9.1 is asked to show below is a question of the first kind, and the paper's criteria receive no confirmation from that dataset in the second, as the disclosure above already said in other words.

2 Two criteria of adequacy

I begin with the two criteria, because they are the sharpest tools in the kit and because their first statements arose in anger — specifically, in my dissatisfaction with superdeterminism, which will serve as the running negative example.

The Strong Implication Criterion

An ontological model which consists with phenomena but does not so much as imply them is itself incomplete (the weak form). The strong form adds: one should not be able to switch the empiricism out for distinct alternatives and fit the base form of the model just as easily. Visually, a model should be wound so tightly to the empiricism that shifting the empiricism from what is known would dissatisfy the model immediately, while not being so hastily thrown together that subtracting empirical knowns would undermine its fit — a cone: sharp focus, wide and thoughtful base.

The strong form is a substitution test, and a substitution test is only as sharp as its three parameters, which the criterion as boxed leaves to the reader: which alternative empiricisms are in play, what counts as preserving the model's base form, and what counts as fitting just as easily. Left open, the criterion can be run after the fact on anything; so it is given a protocol. Let C be a theoretical core, $\mathcal{A}$ an auxiliary class with a stated complexity measure, E the actual empirical structure, and $\mathcal{E}$ a *declared* contrast family of alternative empiricisms, fixed before the test. Define the substitution set $S_C := \{E' \in \mathcal{E} : \exists A' \in \mathcal{A},\ \text{complexity}(A') \leq \text{complexity}(A),\ (C, A') \text{ fits } E'\}$. The strong form then says that an adequate model makes S_C small relative to $\mathcal{E}$: the core with auxiliaries no heavier than the ones actually fielded should fit few of the declared alternatives. A critic can now run the test, and a proponent can now fail it. Superdeterminism is the clean failure case. Suppose a universe in which superdeterminism accurately describes the quantum effect — the Bell correlations implied by an underlying deterministic phenomenon. I then ask why we are to see the quantum phenomenon in the form in which we do at all. Why the wavefunction? Why superpositions? Why the uncertainty principle? Why does any particular quantum effect appear rather than not appear? On the surface this looks like a nitpick; in fact it is the entire question. Superdeterminism does not feign to discriminate between the quantum presentation and a myriad of alternative presentations the same deterministic substrate could equally have worn, and so it fails

the weak form and the strong form together. The positive contrast is instructive: general relativity does not merely permit the perihelion behavior of Mercury — its field equations entail it,[4] so the theory could not survive that behavior's absence, and could not have been refit, unchanged at its base, to a modestly different gravitational phenomenology. That tightness, not mere consistency, is what an ontological proposal owes. The criterion has a near relative in the requirement that good explanations be hard to vary [21], and the kinship is acknowledged; the delta is the strong form's substitution test, which quantifies over alternative empiricisms rather than over variants of the explanation. And one Duhemian caveat is owed [22]: the criterion is assessed against a model together with its auxiliaries, never against a naked core — what must fail under substituted empiricism is the bundle actually fielded. And the delta, per the record, is load-bearing rather than taxonomic. Run as a compass at first contact, the substitution test convicted classical reductionism in real time: a universe in which the quantum reduces *efficiently* to the classical on demand makes no discrimination between quantum-computational speedups being permitted and being forbidden — and the universe is specific, the speedups are permitted — so at least one quantum feature must resist the reduction.[5] That inference arrived during the introductory double-slit video itself, in August 2022: the account stands attested in the 2024 deposit, its statement thereby predating this paper; the dated August 22 entries sit twelve days downstream, the account's "within the week" chaining video to eureka and §9.2's own within-the-week chaining eureka to entry, the two short hops closing cleanly across the gap; and the day-precision, August 10, was read off the author's YouTube watch history in a lookup some two years downstream — a platform-dated log rather than bare recollection, the lookup's report the author's own, with an export or capture of the entry the outstanding step that completes the certificate. Its conclusion is the shape the nonclassicality literature took decades to distill: most of the quantum phenomenology reproducible under classical epistemic restriction, genuine nonclassicality concentrated in what Spekkens has called the "thin film" — the remainder where computational advantage lives [24, 25]. A delta that compresses that distillation into a first sitting is not priced by its sentence length; it is the vacancy column of §7 cashed on this criterion specifically — with the one-way conditional of that section still governing: the yield dates the compass's effect, and the criterion's validity rests, as before, on the external cases alone. As for how a first sitting comes equipped with such a compass, the record's answer runs through the other wing: the author's prior Nexic axiomatization work had trained — on his own reading of his own interior, conjecture-grade like every such reading in this paper, the Grothendieck reconstruction included — the habit of working locally on open questions under the global assumption of a foundation to meet in the middle, gesturing at the outline of a foundation before possessing it concretely. On that reading the criterion's instinct-form is a trained posture arriving before its name — content before name, the record's recurring signature, here on a criterion rather than a construct — with the split graded as everything here is: the practice's priority certifiable in the Nexic wing's dated artifacts, the causal link from practice to instinct introspective, and priced accordingly.

The second criterion needs its target on the page first, because it was built against a specific standing doctrine: *quantum logical antirealism*, the common stance that the quantum world is not merely undescribed but rationally indescribable. The doctrine grades into three orders. First-order: each deeper discovery is intrinsically more impersonal than the last, as if by an ontological trajectory — a falloff between physics and one's tangible life. Second-order: human intuitions

[4]The field equations, and not general covariance alone: covariance constrains the form of a theory, and its physical content has been disputed since Kretschmann [20].

[5]Efficiently: what the speedups exclude is efficient classical simulation, not a classical ontology as such, since ontological models of quantum theory can carry exponential resource costs [23].

and mental capabilities prove insufficient for quantum superposition specifically. Third-order — full-blown — is the Bohr-flavored claim that no tangible logical system can encapsulate quantum effects at all. The load-bearing distinction the doctrine elides is between *a lack of ontological applicability for the standard rational system* and *a lack for any rational system at all*: the former is just the pre-revolutionary condition, while the latter is what the criterion below convicts of paradox. The first order, at least, admits a clean psychological diagnosis: it is the predictable consequence of *a paradigm's epistemology significantly outpacing its ontology*, and nothing more. Had Planck's quanta gone fifteen years without Einstein's ontological interpretation, the construct would have grown mystical to the point of perceived inaccessibility; had gravitational lensing been detected before general relativity existed, it would have read as a defiance of human reason. In each counterfactual the physics is unchanged and the "antirealism" tracks only our temporary lack of ontology — and the species has run the real experiment: the pre-Newtonian planets, pulled along their orbits by invisible angels, were exactly perceived epistemic antirealism, dissolved not by argument but by the arrival of an axiomatic framework. That precedent was my driving intuition at the very beginning: *it has already been done, by Newton himself, so we know at least that it can be done.*

The Strong Rooting Criterion, in two parts

The method (load-bearing): when a standing framework leaves its subject matter unformulable in any human-understood model while the subject matter plainly exists, the conflict is an instruction, not a paradox — abstract to the next layer and relocate the theoretical constraint there. *The existence thesis (contestable, recorded):* a physical circumstance that cannot be formulated in the basis of a human-understood model — "cannot" read in principle, under every extension of human-understood modeling, not merely at present — is not existent but genuinely contradictory.

Two parts, because they carry different weights. Everything this paper does with the criterion — the Copenhagen diagnosis, the re-rooting instruction, the Newton analogy below — consumes only the first part, a discovery heuristic requiring no metaphysics, the toy derivation closing the section excepted. The second part is my metaphysical commitment, stated so it can be contested: it will draw verificationist and idealist charges, and I record my defense in one sentence rather than a section — this is not "shut up and calculate" but a claim about existence, that a universe detachable from its theoretic architecture is not merely unknowable but contradictory, an unphysical human construct disguising itself as a philosophy — while noting that a reader who rejects the thesis outright loses nothing of the method or of what follows from it, save that derivation, whose separability step leans on the thesis's strongest reading.

One compatibility deserves immediate note: the criterion does not conflict with Gödel's incompleteness theorem, because an inability to construct a completely descriptive mathematical system is not the prospect of describing an ontological paradigm — incompleteness constrains the information accessible *within* a paradigm, and says nothing about the ability to establish the paradigm in the first place. Properly denying epistemic antirealism is exactly what lets the counterintuitive logical barriers — incompleteness, uncertainty — be partitioned into their correct ontological stances rather than melted together; the Jaynes omelette, part realities of Nature and part incomplete human information scrambled up together [26], is the standing image of what the failure to partition looks like. The rooting criterion earns its keep through its contrapositive, which functions as a discovery engine, and the engine is best exhibited on the case that mattered to me. The derivational black box of the Copenhagen interpretation fails the criterion — quantum

logical antirealism leaves the quantum circumstance unformulable in any human-understood model — and yet quantum mechanics is physically existent. This conflict yields not a paradox but a direct instruction: abstract the informational context of quantum mechanics to a higher level, and constrain *that* theoretically instead. Upon recognizing that unknowing is a fundamental property of the universe, one ought not stop there and declare the universe logically unfounded, for that is exactly the rooting violation; one must abstract to the next layer and understand that the logical dynamics of the universe do not constrain ignorance versus knowledge — what can be unknown versus known — but rather constrain the *concept* of ignorance: the structural interrelation within ignorance, treating the knowledge of ignorance itself as a kind of knowledge and constraining it relationally.

I take this to be exactly analogous to what Newton realized about motion. Velocity is relative; rather than concluding that the universe has no laws of motion — the rooting violation of his day — Newton inferred that the dynamical constraint attaches not to velocity versus motionlessness but to the abstract study of *change* in velocity. Rooting the constraint one layer up is Newton's second law. The constraint on the dynamics of conceptual ignorance, arrived at by the same move one layer up from the ignorance/knowledge distinction, is the second postulate of the framework developed in the companion manuscript [1], and this is why I regard that postulate as the counterpart in epistemology of what Newton's second law is in motion — a claim I state here and argue there. For present purposes the point is methodological: the criterion did not merely evaluate a finished proposal; run backwards, it dictated where the next theory had to live. A criterion that can do both jobs — filter and compass — is what this paper means by an adequacy criterion worth naming.

One boundary of the criterion deserves stating, because it is where speculation gets its license back. The rooting criterion does not say speculation is unproductive; it says that alluding outward to some vague deeper principle in order to resolve current problems is problematic in excess ("a future version of superdeterminism will resolve the paradoxes"), while establishing accurate broader features of a future theory without alluding to the manner of its completion tends to be healthy ("the issues in current quantum interpretations are categorically like mistakes in Sudoku, pointing to an inconsistency of our own assumptions"). The first outsources the debt; the second maps it.

The two criteria, finally, are not siblings but a lineage, with rooting the more fundamental — and the relation admits what the long manuscript calls a toy derivation, migrated here because the defense of §1.1 leans on relations of exactly this kind. Suppose a universe that violates the implication criterion: superdeterminism as its ontology, quantum mechanics as its empiricism, the fit comfortable. Such a universe has locked in a feature — the theory sits at arm's length from the empiricism, since other empiricisms would fit the same ontology equally well. But that reassignability just is separability of ontology from empiricism, and separability is what the rooting criterion forbids: a physical circumstance whose theoretic architecture can be detached from it is, on the rooting view, not underdescribed but contradictory. Roll the contradiction back and the conclusion follows: a universe violating the implication criterion cannot exist in the first place. A toy derivation, and marked as one — the quantifiers are informal, and the separability step leans on the rooting criterion's strongest reading — but it is a derivation and not an analogy, and it converts this section's two criteria into one commitment viewed at two depths.

3 Installed versus independently entailed structure: the substance/reasoning axis

The substance/reasoning axis

Failed theories tend to add new substance to the universe. Successful theories tend to add new reasoning to the universe.[6]

The historical record sorts along this axis with striking regularity. The substance column: epicycles, caloric, phlogiston, the luminiferous ether — each a new posited *thing*, installed to make the standing reasoning survive contact with the phenomena. The reasoning column: statistical mechanics, which added no new substance to thermodynamics but re-derived it as counting; special relativity, which removed the ether and reorganized simultaneity; natural selection, which added no vital substance and subtracted teleology. On the direction-of-dependence formulation the apparent counterexamples sort themselves: the atomic hypothesis, germ theory, quarks, and the neutrino were all substance *entailed* — atoms by the counting that re-derived thermodynamics, the neutrino by conservation, quarks by the symmetry classification — while epicycles, caloric, and ether were substance *installed*, patches whose entire office was rescuing the standing reasoning from the phenomena. The axis is still not a law; I state it as it behaves: a strong prior with a diagnostic edge. When a proposal's central move is a new primitive, the burden of proof it faces is categorically heavier, and the reason can be given structurally. A new primitive connects to the existing theory-graph by essentially one edge — the introduction itself — and when we look at the actual universe, every last scientific revolution has revealed it to be vastly more interconnected than expected beforehand; a minimally connected proposal is therefore almost surely wrong *as stated*, however good its core idea. I offer that argument as a conjecture with an image rather than a theorem — no measure on theory-space is specified, and I will not pretend one is — but the image has teeth, and the axis it motivates can be checked against the record independently of it.

String theory is the live application, and I want to be exact about what the axis does and does not say there. The arguable core idea — promoting some concept of worldsheet to relevance — is quite clever. But the theory realizes the idea by postulating strings as a new primitive, which is precisely the substance side of the axis, and hence the diagnosis: the proposal as stated is almost surely not what the universe is doing, while the core idea, being good, probably retains shades of similarity (§6) to whatever the universe is actually doing. What I would have recommended instead is keeping the worldsheet idea but evaluating it as an emergent projection from something deeper — a reasoning move — rather than dropping it into the system directly. The axis, note, does not score outcomes by pedigree or difficulty; it scores the *kind* of move, and it makes a falsifiable bet about which kinds age well.

[6] Stated here exactly as the long manuscript states it, because the concision is the point: the axis is meant to be carried in the head, not consulted. The precision gloss, which this section's cases then use: failed theories add substance *installed to rescue standing reasoning*; where successful theories add substance, it is *entailed by the new reasoning* rather than installed to save the old — the direction-of-dependence formulation on which the apparent counterexamples sort themselves. The gloss is the axis's real content, and it deserves the name that says so: *installed structure* against *independently entailed structure*. Without an operational reading the taxonomy is retrospective—every successful posit becomes "reasoning" by saying it was entailed, every failed one "substance" by saying it was installed—so the temporal order is made the criterion. A posit P is *independently entailed* if the constraints from which P follows were fixed before, or independently of, the anomaly D it addresses; P is *installed* if its only evidential role is introduced after D, specifically to absorb D's residual, while adding no independently exposed consequence. Read that way the axis is decidable from dated records, which is what makes the dataset of §9.1 bear on the criterion itself rather than only illustrate it.

The axis has ancestors, and naming them is how its delta gets measured. The nearest is Einstein's own 1919 distinction between principle theories and constructive theories [27] — thermodynamics and relativity as theories of principle, kinetic theory as construction from posited constituents. The kinship is real, but the claims differ in force: Einstein's was a taxonomy of forms; the present axis is a *diagnostic* — substance-addition as a predictor of failure — which the 1919 taxonomy never asserted and which the constructive successes (kinetic theory itself) show must be probabilistic rather than exceptionless. The second neighbor is the distinction between progressive and degenerating research programmes [28], which tracks a programme's trajectory over time by whether its adjustments predict novel facts; the present axis is instead read off a single proposal's structure at birth, before any trajectory exists — a prior where the programmes machinery is a posterior. Between the taxonomy that does not predict and the tracker that must wait, an axis that bets early and states what would refute it has, I think, its own honest work to do.

4 Kinds of revolution: the ontological and the methodological

The standard chronology of physics is a chronology of unifications, and I think its implicit goal is to categorize Newton and Maxwell the same way. As a heuristic for *outcomes* that is solid enough; as an account of *kinds* it misinterprets the record. When Faraday came out with his experiments and Maxwell went to write them up as mathematics, the task itself was of the shape "carry the surprises unveiled by the experiments to axioms" — fundamentally important, and additive. What Newton did was of a different shape and different stakes: "perform a titanic-level unscrambling in our own priors as humans, which has been stuck at an impasse since the Stone Age" — much more subtractive by weight than it was ever additive. To give the kinds names: Maxwell's is a methodological-conceptual revolution; Newton's is a categorically *ontological* one. The distinction matters now because contemporary quantum foundations has been issued exactly the Newton-type assignment — the call to unscramble within our own fundamental conception of causation and inference [29] — and mistaking it for a Maxwell-type task is a way of guaranteeing it never gets done. One positioning paragraph against the obvious neighbor: the kinds distinction cross-cuts, rather than restates, the normal/revolutionary structure of [30] — Maxwell's achievement was fully revolutionary in Kuhn's sense while methodological in mine, so the present distinction sorts *among* revolutions by the character of the central move, where paradigm talk sorts revolutions from puzzle-solving; I use the word paradigm throughout in Kuhn's debt and with that difference flagged.

So that the kinds do not collapse into a headcount, a criterion independent of authorship: a revolution is *ontological* exactly when its central move re-identifies what the theory's basic quantities are about — an unscrambling of the referents in our own priors — rather than extending, formalizing, or systematizing a referent already in place. By that criterion Newton and Einstein[7] classify as ontological and Maxwell as methodological on the moves themselves, with no one

[7]Another interesting note I want to call out is that with particularly profound conceptual shifts, in general they tend to be characterizable as promoting to the very primitive of the ontological apparatus something which in the previous framework was a deduced identity. The canonical instance, perhaps, is Einstein's equivalence principle rendering the local character of free fall as inertial transforming a deduction from gravitational force to the root intrinsic property of space around which the new theory of gravity inferentially assembles.

In my own work on Aethic reasoning, once I instated the conceptual shift that is the relativity of determinacy [31, 1] by means of the extrusion principle [1], I then inferred upon completing the postulates that the Quantum Nexus and Powerset Invalidation Theorem must be in effect for consistency with quantum wave mechanics and the Born rule [1], which in turn deduced the quantitative wave-particle duality relations and no-signaling theorem prior to the quantum wavefunction rather than after it [32, 1].

counted. My base understanding of the surrounding sociology is Kuhnian [30]: the individual and the collective are both absolutely indispensable to the flow of progress — individuals for sparking paradigm shifts, the collective for the normal science between them — and the great-man-versus-collective debate borders on naïve in insisting we pick one; the body depends on both the heart and the spine, and removing either kills instantly. Any appeal to the historical record of "great individuals" also carries an obligation stated plainly: that record is severely compromised by the systematic exclusion of women, people of color, and other marginalized groups from education and scientific institutions — an exclusion that speaks only to the cruelty of the structures enforcing it and says nothing whatsoever about the capacities of those it targeted, the conflation of denied access with diminished ability being among the most damaging errors in the history of ideas. The irony is particularly bitter[8] when you also consider how the malice behind historical systemic discrimination muzzles the very class of ontology-capable individuals we should be prizing — while the very system doing the muzzling insists that individuals are special.

Where I differ from the base Kuhnian account is one attribution. The record then suggests a demographic regularity, stated as the conjecture it is rather than as a historical result: that the ontological move has, historically, been formulated by a single mind. The claim rests on four individuations a specialist historian is entitled to contest before granting it—what individuates an "ontological move"; why Newton and not Newton with Huygens, Hooke, and Leibniz; why Einstein and not Einstein with Lorentz, Poincaré, and Grossmann; and whether "single mind" is partly manufactured by naming conventions and retrospective attribution—and until those are settled on a corpus not selected by outcome, the regularity is a conjecture the cases illustrate. It neither ranks the individual above the collective nor romanticizes anything; it identifies the one feature of progress that appears individually necessary while everything else is collectively performed. Quantum mechanics is the test case, and the criterion lets it fall where it falls: collectively built, magnificently systematized — the Dirac equation easily as elegant as special relativity — and, on the referent question, still open in the precise sense that no candidate re-identification has been collectively ratified (Everettians and Bohmians claim the move completed, without commanding ratification), while the causal-inferential program [29] treats the unscrambling of causation and inference that would constitute quantum theory's ontological move as the outstanding task, not a completed collective achievement. The regularity thus survives without fiat: the collective built everything about quantum mechanics except the one move the criterion tracks, and no performed candidate for that move has been collectively ratified.

On the inevitability thesis [33] I will be as delicate as I can while still saying it: I think special relativity, and general relativity above all, would have been quite a long shot without their author — the "thin film" of ontology is, in my experience, overestimated in prospect and underestimated in retrospect. Everything may be in the air empirically, as the measurement problem has been since at least the 1940s; but Poincaré was brilliant and ontology was not his thing, and if the collective's access to tiny ontological perturbations were what it is assumed to be, some anthropic reasoner would have reached the Accordance Principle [34] long ago. The collective finds such perturbations far more inaccessible than is typically assumed, and living amid the further revolutions none of us have achieved is, for the inevitability thesis, like living the refutation of one's thesis without noticing it. The section compresses to one principle, stated in the source development and adopted here as this paper's motto for the ontological kind: ***scientific progress is only ever as good as its ontological content***.

[8]This disconnect is one of those rare things that actually brings me physical rage: the juxtaposition of denying individuals gratitude because of race or gender with then targeting those same people for reeling under that discrimination — for struggling to reproduce the output stats of the white men those same institutions babied.

There is a further asymmetry worth naming, because it explains why the ontological kind is so rare outside physics. Distinguish a concept's *intrinsic* advancedness from its *perceived* advancedness — the latter read off our own paradigmatic orientation, the former off something closer to inferential fruitfulness. My proposal is that philosophy as a discipline carries an abnormally high ratio of perceived to intrinsic advancedness, and physics the lowest, with the human-nuance-inviting disciplines grading between. The reason is that philosophy is perpetually stuck in a paradigm of terminology where trivial statements are arbitrarily hard to even state and standing unifications go unrecognized — a diagnosis with a natural ally in the analysis of merely verbal disputes [35] — a state I would compare directly to physics before Newton, or, at the frontier, to quantum field theory today, where one senses a clean partitioning in the universe's own contributions while our drawn borders intersect them so chaotically that it is impossible to parse how much of what we write is ontological signal and how much is the noise of human convention. The difference between science and philosophy is then that each new scientific revolution is low-hanging fruit compared to a philosophical revolution of comparable intrinsic advancedness: the scientific case has an ontological core underneath, which provides just enough leverage that a Newton-caliber mind alone can access it. The compounding difficulty as one moves away from convention-impartial ontology is, I will suggest in §6, not incidental — it is the lexicon itself doing the scrambling.

5 Two stages of a breakthrough: charming and structural

The two-stage model

Inference-driven breakthroughs tend to unfold in two stages: a *charming stage*, in which the researcher has intuited a basic principle of the universe but still views it from within a subtly anthropocentric vantage; and a culminating *structural stage*, from which the charming stage reveals itself as having been charmingly naive about the tightness with which the universe actually does things.

Where the universe does things like an all-business program director — every move a symphony in ontological graph-connective contribution — a human first gaining a partial copy of the memo tends to situate the plan in terms of what is familiar, and that is where the retrospective naivety comes from. Two historical cases clued me into the phenomenon. Feynman's "Plenty of Room at the Bottom" lecture [36] initiates squarely in the charming stage: the foundational idea of nanotechnology intuited — retrospectively discussed as prophetic — while the concrete proposal of iterative hands proved unnecessary to the prizes themselves, the first won unexpectedly quickly by classically available means, the second later by a more streamlined method matching the idea while stripping the flanks. And Einstein's road to general relativity reads as the model's cleanest instance: the equivalence principle as the charming stage — the principle intuited, still situated in elevators and observers — and the transition to the structural stage arriving with general covariance, where the familiar scaffolding fell away and the constraint stood alone.

The model's most distinctive prediction is also its least testable from public records: the charming-stage thinker is surprised by the developments of *their own future self* — just as one cannot predict the specifics of a future technology even while getting the pattern extraordinarily right, so the structural stage outruns what its own author would have forecast months earlier. Testing that requires longitudinal data on a single mind, dated finely enough to catch the surprise in the act. That is precisely the kind of data §9 supplies — and one further move, available to this methodology alone, closes the circularity that would otherwise haunt it: *date the theses*

themselves. The theses of this paper were plausibly formed in part from the very record they are run on; but the record's deposits are frozen, so whether a thesis was stated *before* a transition it retrodicts is a lookup, not a claim. In particular, if the two-stage model's statement stands in the frozen November 2024 print, then the 2024→2026 transition — the naming layer surprised by the formalization layer — is a retrodiction out of sample in its dating, though not in its coding;[9] the check is a recomputation against version 1 and is routed, with the other certificates, to independent audit.

6 Shades of similarity, and why deep unknowns hide

Shades of similarity

The deepest unknowns tend to have shades of similarity to all things known. It is hard to discover them not because they are new, but because they resemble conflicting sides of existing debates.

The mechanism is camouflage. A deep unknown that resembled nothing would at least register as an anomaly; one that resembles *both* sides of a standing debate gets claimed by each side as a flavor of what it already holds and dismissed by each side as a flavor of what it already rejects — and so it hides in plain sight, misfiled rather than missed. The thesis makes a standing prediction about reception that I state here as a falsifiable expectation for my own framework: readers will systematically classify it as a variant of whichever position they already occupy, and the misclassifications will be confident, articulate, and mutually inconsistent. Stated bare, that prediction doubles as armor — any criticism files as confirmation, the conventionalist stratagem this paper's own citations condemn — so here is the exit condition that immunizes the immunization. A *correct* classification is one that locates the framework by its own stated deltas: each companion paper carries an explicit statement of where it diverges from its nearest neighbors, and a reception that names those deltas — agrees or disagrees with them, but names them — scores as correct. The prediction is scored on aggregate published reception tallied against those delta statements, and it *fails* if the modal published response locates the deltas accurately; camouflage, on this scoring, is a measurable property of a literature's response, not a universal escape hatch.

I now want to argue that this camouflage operates at a level deeper than concepts: it is built into the words. Take a tree model of cognition — cognitive branches, each with its own set of intuitions, one branch having happened to write the public lexicon. (The documented instance is the allistic/autistic case, where the diagnostic binary inadvertently classifies people as "non-allistic" for lacking one branch's traits, on the direct analogy of the pre-Darwinian human/animal dichotomy against the post-Darwinian tree; but the argument runs on branches generally, and nothing below depends on the clinical vocabulary.) Consider the standard characterization of mathematics as "logical" — right and wrong answers — against the humanities as "nuanced." Anyone learning the most introductory layer of a topic tailored to a different branch's intuitive landscape reports the same characterization: kindergarten literature has clear rules — either the letter is written this way or it is not — and sits at an isomorphic introductory level to twelfth-grade mathematics. Comparing twelfth-grade mathematics to twelfth-grade literature is therefore the wrong reference class; the felt contrast measures the general population's higher intuitive capacity

[9] Whether a transition shows the two stages is still read by the thesis's author, so the retrodiction is out of sample in its coding only if that reading is fixed before the transition is known; the companion audit's markers, fixed afterwards, have not been applied to it [2].

for literature, not the topics. Had the branch with the flipped capacities written the linguistic rules, mathematics would be described as highly nuanced and flexible — as it already is by the world's great mathematicians — and Shakespearean literature as a rigid system of writing down letters and sounding them out. Symmetry under re-choice of reference branch is the sign of a good universalization, and this vocabulary has none.

Call the resulting phenomenon *linguistic masking*: the use of dominant-branch-perspectival vocabulary to describe one's own internal cognitive phenomenology. The phenomenon is proposed here as a general one, and it is stated as a hypothesis with its test attached rather than as an observation that establishes itself: give blinded readers semantically novel theses whose surface language closely resembles established positions, with the delta explicitly present in the text, and measure whether they systematically assimilate the thesis to the nearest familiar category anyway. A single reader's misparse, human or machine, is an instance generating the hypothesis; the hypothesis is confirmed or not by that experiment, and nothing below leans on its truth. The public cognitive lexicon is not a neutral medium but a branch-specific dialect that other branches must translate themselves into, often by silently redefining the available terms while preserving their public surface — producing semantic drift the speakers do not notice, or leaving labels standing with subtleties behind them that no other branch is aware were arrived at. The person who multiplies eight-digit numbers by intuitive nuance will still call mathematics "rigid and logical," mirroring where the dominant branch applies the word rather than reporting their own calibration — and the labor is asymmetric and continuous, extracted by the standing lexicon whether or not anyone is presently misreading you.

The cleanest exhibit on record is Grothendieck's famous dislike of examples [37, 38] — with the reading that follows marked plainly as my conjecture about his interior, not as biography. From within his historically rare cognitive branch there are plenty of things semantically consistent with examples — picture an order-five abstraction grasped by how three order-four objects fit together in various ways: each snapshot shows a facet of a deeper thing in a targeted simplified context, which *is* the semantics of an example — but only Grothendieck could recognize such an object as one, since nobody else gets past a single order-three object in a frame of consciousness. What everyone else definitionally categorizes as "examples" is the same construction run over order-two objects, and *that* category of instances — not the semantic concept — is what Grothendieck reports disliking. The mismatch then locks from both directions: he assigns his own semantic character to the dominant branch's categorized instances, they read his sentence under their semantics, his biographers explain an eccentricity he never had, and the concept he actually valued — his own branch's example — likely never gets written down at all, occupying as it does the base of his cognition beneath everything he names. Deep unknowns hide behind conceptual resemblance; deep *minds* hide behind lexical resemblance; and the second mechanism feeds the first, since the unifications the lexicon scrambles are exactly the ones §4 says philosophy fails to recognize.

The move that generates such reframes deserves its own name, because it is not local to cognition: call it *localizing the flex to the environment*. Confronted with the fact that apes do not ask questions [39, 40], do not conclude a missing faculty; instead import full human categorical cognition into the ape unchanged, and ask what their contextual environment would have to look like to reproduce the no-questions behavior for consistency — whereupon the answer arrives as a boundary-prediction difficulty compounded by language funneled through blinders, and an ape asking a question is priced correctly: as historically anomalous as Sequoyah, illiterate, deriving the Cherokee syllabary by inference alone. This is, in retrospect, exactly the move that gave rise to Einstein's equivalence principle — one of the highest-caliber scientific ideas in known history, and methodologically informative for that reason: hold Earth's phenomenological setting

constant, import the accelerating-laboratory empiricism fixed, and let Earth's structural-contextual environment vary for consistency; the result is that spacetime must be curved here. The reframe and the principle are one method — keep the interior constant, flex the context — and it is the method by which a dominant branch's deficit-readings of other minds get replaced by structure.

One more unification, because it converts the whole discussion into a prediction problem. Communicating across a cognition branch and predicting the pinpoint structures of a future society are, I claim, the same difficulty rather than categorically different ones: design a Turing-style test in which an objective actor must distinguish a communication across a paradigm boundary from a one-way communication aimed at an unknown future society, and the actor cannot distinguish them on categorical grounds — each could be labeled as the other while remaining consistent. So whoever wishes to price the true difficulty of crossing any such boundary, without falling back on heuristics, should simply ask how difficult it would be to predict the pinpoint nuances of the societal structures thirty years from now. The camouflage thesis, the lexicon thesis, and the boundary-prediction unification are one phenomenon at three depths: the unknown resembles the known, the words belong to one branch, and the far side of any deep boundary is exactly as unforecastable as the future.

7 Pricing a contribution: proximity, vacancy, and the co-signature loop

The preceding section explains why deep unknowns hide. This one draws the consequence for evaluation, and it inverts a heuristic evaluators use daily: shades of similarity make *short* gaps disproportionately hard to close. A position crowded by near neighbors reads as occupied from every direction — the density of the neighborhood is mistaken for coverage of the point — so the closer the standing debate presses around an unoccupied bullseye, the more certainly each camp files the vacancy as territory it already holds, and the harder it becomes for anyone to see that the decisive spot is empty. The firewall debate of §1 is the worked instance, and its standoff is external history: for some eight decades the discovery/justification argument ran between a camp that discarded the context and a camp that hunted rules for it, while the split position — no rules, full evidential retention — sat between them resembling both and claimed by neither camp; a middle that near is precisely what this section predicts goes unoccupied longest.[10] Two disciplines follow. For evaluation: when a contribution's deltas read as small against a dense neighborhood, the evidence of its difficulty is not the size of the textual delta but the duration of the vacancy under that density — similarity is a fact about the terrain's crowding, and down-weighting novelty by resemblance systematically penalizes exactly the contributions this section predicts are hardest to make. The correction is not a sign-flip: "close means low significance" is the fallacy, and "close means high significance" would be its equally naive mirror — worse than naive, in fact, since a proximity-scoring rule would attempt to axiomatize a leg of inductive evaluation, exactly the move §8 argues cannot be made. The asymmetry is worth one more sentence, because it locates the line exactly: a positive comparative is monotone — affirm "closer is harder" as a principle and you have asserted "farther is easier" in the same breath, a commitment nobody holds — while the negative form, *do not assume closer is easier*, asserts no ordering and denies two: it

[10] Unoccupied by the camps, though not unstated: Hoyningen-Huene, cataloguing the distinction's versions, located its defensible core in the difference between the descriptive and the normative perspectives on science [41, 42], which admits the facts of discovery to description and reserves appraisal for the normative perspective. That discharges the claim that the position went unstated, by the rule stated below; which of §7.1's bins the instance then falls in turns on how the camps engaged that statement, which is not assessed here.

calls false "closer is universally easier" and calls false "farther is universally easier" alike — the two universals the fallacy and its mirror respectively require — leaving the terrain order-free, structure removed and none installed, which is the only shape of guidance an art admits. The two condemned universals even co-sign a shared assumption — that difficulty is a monotone function of proximity at all — and the tip is the cut along exactly that co-signature boundary: the corollary's own machinery, run one level down on its own evaluation rule. Negatives of that species sit safely on the tips side of §8's line that heuristics are not axioms, the register in which writing keeps Vonnegut's eight rules and remains an art. The correction is an added *column*: the full price of a contribution is the delta statement together with a vacancy report — duration and density, the latter obliged to name the currency it is counted in, since closeness measured in shared formal apparatus can be spurious where the missing piece is ontological (§7.1) — with one discriminator carrying the weight: an *unstated* vacancy is what camouflage predicts, while a *stated-and-rejected* one leaves tracks and counts against the closer — refined at §7.1, where a first run forces a third bin, *stated-but-unengaged*, which is masking rather than rejection, and narrows the second to rejections that engaged the version later accepted. A claim that a position went unstated is written to be defeated by exactly such a track: a critic who produces a prior statement of the position has not merely scored a point but discharged the claim, and the corollary tells them so in advance. And for honesty, the conditional runs one way only: proximity-camouflage explains, *given* a proposal's correctness, why its gap stayed open and why its neighbors miss something decisive for a subset of applications; it is not itself evidence of correctness, and nothing in this corollary upgrades resemblance into support — the scoring rule above remains the only cash this thesis pays.

The corollary also has a dynamic form, and the dynamics are worse than the statics. A fierce two-sided debate is not a neutral search over its question: each camp's advocacy promotes everything the camps share — their shared-right and their shared-wrong alike — because the fight concentrates all critical attention on the contested axis while the common background rides along co-signed by both sides at once, gaining apparent solidity with every round precisely because no round examines it. Call this the *co-signature loop*. Closing the gap then requires something no amount of further fighting supplies: cutting the possibility space exactly along the boundary between what the camps share rightly and what they share wrongly — keep the first, discard the second — and that cut is excruciating from within the paradigm, first because shared assumptions are invisible as commitments (nothing in the debate marks them as claims at all), and second because the loop has been compounding their credibility for decades. The firewall instance now reads one level deeper: what the two camps co-signed was the coupling of the discovery context's evidential status to its normative status — take discovery seriously and seek rules for it, or refuse rules and discard the records — and the split position is precisely the cut along the co-signature boundary, keeping each camp's right half while discarding the assumption both were fighting on top of. Stated as an image rather than a theorem, in the manner of §3's graph conjecture: a debate is a one-dimensional contested axis through a very high-dimensional possibility space; scrutiny flows along the axis; the orthogonal complement — where the shared-wrong lives — receives co-signature instead of scrutiny; and the cut that closes the gap is a hyperplane the debate's own gradient never points at. Which is why prolonged fierceness and prolonged vacancy are found together.

The corollary complex has one standing relative that deserves its name here: the problem of unconceived alternatives [43]. Stanford's historical case series establishes that working scientists systematically fail to conceive of alternatives that later inquiry finds compelling, and deploys the fact as a new induction against scientific realism. The relation is a division of labor with the

delta stated at both ends: his record establishes *that* alternatives go unconceived; the corollary and the loop propose *which* and *why* — the near ones, because density reads as coverage, and compounded, because fierce debate credentials the co-signed background — while the vacancy column converts the diagnosis into an evaluation practice rather than an argument about realism, on which nothing here takes a side. The engagement also yields a check that someone else can run: if the mechanism is right, Stanford's unconceived alternatives should cluster in the neighborhoods their eras' fiercest debates crowded most densely, and his own published cases are the natural test bed — a retrodiction stated here in advance of anyone running it.

7.1 A first run of the retrodiction

The check just stated was run once, in the session that installed it, and is reported here with its results and its corrections — two of which fall against the corollary rather than for it.

The two accounts are complementary rather than rival, and that is what makes a check possible at all. Stanford's own proposed mechanism is cognitive-general: he suggests that the failure to consider relevant alternatives in theoretical contexts may be a general property of human cognition [43]. The corollary proposes a structural amplifier on top of whatever that baseline rate is — and the two differ about *location*. A purely cognitive-general account predicts no particular relation between where a vacancy sits and how its era's debate was structured; the co-signature loop predicts the vacancy sits in the co-signed complement rather than anywhere on the contested axis.

Protocol, with its weakness stated first: for each case, identify the era's fiercest debate in the domain, identify the alternative later communities accepted, and code whether that alternative required negating an assumption both camps held — or instead occupied an open position on the contested axis, which counts as a miss. The weakness is that the cases were selected and coded by the corollary's own proponents; this is exactly the confirmation structure this paper's protocol flags on anyone else, and it is flagged here.

Stanford's core series — Darwin's pangenesis, Galton's stirp theory, Weismann's germ-plasm [44, 45, 46] — codes as a fit. The fiercest contemporaneous axis in that domain was the inheritance of acquired characters: whether the soma writes to the germ line. What all three theories co-signed across that divide is that heredity and development are one problem solved by a material representing the organism's parts, development being the parceling out of that representative material — gemmules thrown off by the parts, stirp elements standing for them, determinants distributed differentially as cells specialize. The alternative later communities accepted negates the co-signature rather than either camp's position: heredity as combinatorial rules over discrete factors that segregate, and differentiation as differential *expression* of a complement every cell retains rather than differential *distribution* of a complement every cell spends. That negation arrived in two stages separated by half a century — segregation with Mendel's rediscovery in 1900, the retained-complement picture only with mid-twentieth-century work on differentiation — and the coding above credits the co-signature's defeat to the pair, not to either alone. No further answer about whether the soma writes to the germ line — no position on the contested axis — is what closed the gap. One qualification belongs with that coding, and it cuts: the axis was fiercely contested mainly in the 1880s and 1890s, Weismann against the neo-Lamarckians, whereas at the Darwin end of the window the shared assumption sat nearer consensus than contest. The series therefore supports the statics — an unexamined shared background — across its whole length, and supports the dynamics specifically, where a live fight credentials that background, only in its later window.

Four further cases, drawn from the standard canon rather than from Stanford's own case

studies — of these, the ether and caloric are named among his book's background examples, while light and drift are this paper's additions — code as follows. Light: the wave–corpuscle debate co-signed both an exclusive disjunction and the demand for a mechanical substrate, and the accepted alternative denied the disjunction. The electrodynamics of moving bodies: the contending ether models co-signed absolute simultaneity, and the accepted alternative denied it. This case is usually told as the proximity claim's most famous instance — the nearest neighbours, holding the transformations already, being precisely who did not close it — and the telling needs a correction that cuts against this section's own convenience. The standard historiography treats the pre-1905 work on the dynamics of the electron as supplying most of the structure of special relativity, with the ontological completion as the piece still missing [47, 48]; that framing prices the ontology as one more brick for the wall. The corpus's own treatment rejects the pricing [49]: the ontological substrate is not a brick but the tree the mathematical pieces hang on — remove it and what is left is not a nearly finished theory but broken glass on the floor — and that substrate is itself the long sequential chain rather than the short step left over. Two consequences follow for the machinery of this section, both restrictive. Apparent proximity counted in shared formal apparatus systematically overstates closeness wherever the vacancy is ontological, so a vacancy report must declare the currency its density is counted in. And the impression that such a result was "in the air" is partly manufactured by its own arrival: a field can circle a problem for a century, and an early landing then makes the whole approach look as though it had been nearly complete all along — with the possession of every loose mathematical piece being at once the reason the thing looks solvable and the reason it stays unsolved. Continental drift: fixism and drift co-signed a static ocean floor, and the accepted alternative moved the floor. Heat codes as a partial fit only, and is reported as such: the kinetic camp did win its axis, and the co-signed background — heat as a classical-mechanical property throughout — failed later and elsewhere.

One finding arrived unbidden, on a different thesis. Coded on §3, Stanford's series sorts cleanly: each failed theory adds *substance* — gemmules, stirp elements, determinants — installed to keep standing reasoning about representative inheritance intact, while the accepted alternative adds *reasoning*, its discrete factors entailed by the combinatorial rules the ratios demanded rather than installed to rescue anything. That is the axis running correctly on a case series someone else selected for an unrelated purpose, which is the only kind of external test that costs its author nothing to fail.

Four corrections are forced, against material installed earlier in this section and against the run itself. First, the vacancy discriminator's two bins are insufficient. Mendel's paper stood in print from 1866 and functioned as unconceived until 1900: a *stated-but-unengaged* vacancy, which is masking rather than rejection, and which the two-bin rule would have miscounted as evidence against the closer. Drift forces a further split: it was stated and rejected for a real reason — no mechanism — and the vacancy persisted anyway for the modified version that later won, so a rejection can be correct on the stated form while the gap stays open. The rule that a stated-and-rejected vacancy counts against the closer is accordingly weakened to this: it counts against only where the rejection engaged the version later accepted. Second, the run does not test the retrodiction as stated. That prediction was distributional — vacancies *cluster* where debates crowded densest — and clustering requires a contrast class this run never assembled: dense debates that produced no vacancy, and vacancies arising where no dense debate stood. What was actually tested is the conditional question, given a vacancy, whether it sat in the co-signed complement. That is worth knowing and it is a different claim; the distributional version remains untested, and §7's statement of it should be read accordingly. Third, every case examined is a famous filled vacancy, which is selection on the outcome, and the run returned no misses at all

— a result to be read as evidence about the coder before it is read as evidence about the world. Two candidate misses were not tested and belong in any independent run: phlogiston to oxygen, where the accepted alternative arguably occupied the contested axis rather than negating it, the co-signed premise of a transferred principle surviving the reversal of its direction; and miasma to germ theory, where the contagionist position stood on the axis throughout. Fourth, and without softening: one self-coded run over partly familiar cases is not a result. It is a pilot.

What would make it a result is specified here in advance, so that the specification is dated rather than fitted. Fix the case list to Stanford's three published case studies and add none. Have the debate-density and co-signature coding done by someone with no stake in the corollary, working from primary sources rather than from this description. Count as a miss any case where the later-accepted alternative occupies a position on the contested axis instead of negating a co-signed assumption. The corollary predicts fits in most; a cognitive-general account alone predicts no such concentration. Any independent run should also engage the critical literature on this same series, which has been re-analyzed with the argument that biologists avoid this inferential pattern where they can and hedge where they cannot [50] — a challenge to the case study's framing that bears directly on how every case above should be coded.

8 Induction as an art form

The art thesis

Inductive reasoning is an art form, while deductive reasoning is part of the artwork. And: inductive reasoning is the art of editing your own faulty assumptions from new data.

The thesis is stated at its defensible strength, which is conditional and not categorical. What the negative argument establishes is that current deep discovery appears to require open-ended representational revision that no fixed deductive or inductive algorithm captures; the word "art" names that feature, and does not prove its irreducibility. Machine discovery, Bayesian experimental design, automated conjecture generation, program synthesis, and evolutionary search all press on a categorical reading, and none of them presently reproduces the reframing this paper cares about; the exit condition below—revise the thesis if a machine independently performs the reframing—is the discipline that keeps the claim conditional now rather than after the fact.

Because so much of the medium of science is nomological, deductive reasoning becomes perhaps its central algorithmic feature — the conversion of first principles into applications, utterly central to scientific output at least as far back as Newton. But that is on the completed-artwork side, and tells you nothing about how to produce the artwork. Good writing is not good because it follows a set of rules, and good visual art is not good because it follows a set of rules; the best available guidance for inductive practice is therefore of the kind of Vonnegut's eight rules for writing fiction [51] — decentralized tips on how flavors pop up across good practice, operating in a heavily flexible medium, and famously issued alongside the permission of great writers to break every one of them.

The argument for the thesis, rather than the analogy, is a regress. Suppose one declared deductive first principles for good inductive reasoning. Induction's entire office — per the second box — is the dethroning of currently faulty assumptions; axiomatizing the practice would smuggle the era's assumptions into the axioms themselves, defeating the purpose the practice exists for. The way an assumption gets dethroned is flexibility and visibility into loopholes, which is why the working posture is the black-hatted outlaw who games vulnerabilities, not the sheriff who follows every rule — and why, when I see somebody invent the axioms of inductive reasoning, what I

see is somebody who has spent a lifetime inside the finished artwork and thinks that experience transfers to the artwork-making. The thesis owes an answer to its strongest opponent, and the opponent has a name. Herbert Simon spent a career arguing that scientific discovery has a logic [52], and built, with colleagues, programs that rediscovered Kepler's third law from data [53]; contemporary machine-assisted conjecture generation renews the pressure — and this very paper documents an AI-assisted process while asserting there are no rules of discovery. The resolution is the line the Vonnegut framing already draws: *heuristics are not axioms*. The regress argument targets axiomatization — fixed first principles that would freeze the current assumptions into the method — and Simon's own machinery is on the near side of that line: heuristic search under human-prepared representations, tips operating in a flexible medium at scale, artwork-side pattern extraction that presupposes rather than performs the reframing this paper calls the making. On the same line, the corpus's own drafting collaboration classifies cleanly: the machine amplified — searched, formalized, verified, pressed objections — while every reframing ruling ran through the author's taste, and no rule set generated the rulings; the outlaw kept the wheel, the sheriff got a faster horse. Should a machine one day perform the reframing itself, unprompted, the thesis would owe revision — I state that as the thesis's own exit condition rather than defend against it in advance.

Note carefully what the thesis does *not* say: it places no limit on rigor after the making. The firewall position of §1 is exactly this division of labor — no rules of discovery, exacting rules of checking — and the two postures work different shifts on the same job.

The thesis also settles a standing misreading. "Inductive reasoning carries special cases to general cases" is correct as a retrospective and misinformative as a procedure — and that procedural misinformation is exactly where a familiar category error breeds: the resistance of inductive practice to axiomatization gets read as an open problem, when it is simply the categorical nature of an art form. This is the calibration in which I engage the classic confirmation paradoxes [54, 55]. Those puzzles arose within and against the formal-confirmation program — Hempel's ravens as an internal puzzle of confirmation theory, Goodman's grue against its syntactic core — and the positive conclusion their best author reached — projection settled by entrenched practice rather than by logic — already concedes much of the negative point; my target is the surrounding reception, the idea that grue creates a foundational crisis in how to do inductive reasoning, as if axiomatic inductive reasoning were a live option rather than the thing the regress argument above rules out. In actual practice it is a question of taste — exactly as in art — whether a perturbation to a concept is to be entertained; no heuristic for when to make those leaps would be honest, because the freedom to work the shape of what is needed by feel is what categorically makes success possible at all.

Last, the conjecture that says why the medium is as messy as the thesis requires — offered, per this paper's discipline, as a conjecture with an image. Take any deep standing model and let its complexity divide into proportion α human bias and proportion $1 - \alpha$ ontological nomology. Deduction operates entirely in the $1 - \alpha$ subspace — all nomology-to-application carrying happens there — and yet, measuring dimensional complexity, the universe does things simply relative to the accumulation of human biases and misinterpretations over the millennia: $1 - \alpha$ is measure zero with respect to α, as if comparing the fractal dimension of the directional field of Einsteinian gravity against that of a brain. No measure is specified and I will not pretend otherwise; but if anything like the image holds, then the ability for inductive reasoning is the ability to parse the α-measure subspace and compress it into something transferable to the $1 - \alpha$ part — work that must be done scrappily, by feel, by artists by trade, because knowing what to expect ahead of time is exactly the luxury that α's existence denies.

A corollary on mathematical beauty closes the section, because the art thesis is often mistaken for the Platonic one and the difference is exact. Mathematical beauty is a subjective concept — two people rate the same concepts differently — and unaided human guessing over the space of *arbitrary formal statements* succeeds at an infinitesimal hit rate that fails exponentially faster under compounding criteria; the reference class matters, since trained conjecture over structured domains does far better — Ramanujan is the standing existence proof — and the argument below concerns the former class, where the beauty signal has to do its work. What beauty supplies is therefore not truth but *non-mediocrity*: since truth necessarily fails probabilistic mediocrity inside standard human intuitive capacities, only what stands out at an unusual quantile is even eligible, and the startling character that subjectively marks a formalism as beautiful is precisely what clears that hurdle. Beauty and truth are thus strongly correlated without either causing the other: beauty buys a candidate its *chance*, generally still far from likely. This also positions the corollary beside its near-ally: the argument that beauty has led physics astray [56] is, on the present account, half right — beauty never was a truth-maker, and treating eligibility as likelihood is exactly the error; what the corollary preserves is beauty's legitimate office as a non-mediocrity filter, nothing more. The working practice this licenses is augmentation rather than divination. Concretely: needing a y that meets three criteria while direct intuition supports only two, set $y = \sin(2 \arctan x)$ for an intermediate x fitted to the two — whereupon the third criterion, boundedness, arrives free from the identity web itself ($\sin(2 \arctan x) = 2x/(1 + x^2)$, confined to $[-1, 1]$ by construction) — the identity web of trigonometry, whose internal structure outguesses any human; the discipline's own elegance, not the practitioner's eye, is what reaches the final gap.

The thesis has a corollary in the sociology of evaluation, worth stating with the same one-way discipline as everything else here. A metric that could *proactively* and reliably identify frontier inductive capacity — score the genius before the output — would have to span the space of moves it claims to score; a scorer that reliable is a generator in disguise, and its existence would constitute exactly the rules of discovery this essay denies. The direction matters: verifying a *delivered* result is often cheap, and that asymmetry is the whole economy of proof, but scoring an *undelivered* one requires modeling the prover, and no such model is on offer — our best statistical evaluators, whatever else they do, are not emitting theories of everything. The corpus's own outlier-productivity study sharpens this asymmetry rather than softening it [49]: capacity of the relevant kind is measurable *retrospectively*, from achieved sequences and the step counts they required, and the measurement returns extreme values — which is exactly what makes the proactive case hopeless, since the retrospective score is computed from the very output whose existence a proactive score would have had to anticipate. So the fallibility of genius-metrics is not an empirical embarrassment awaiting reform but an entailment of the art thesis, and its signature failure mode follows: the rare uncredentialed person significantly outpacing the credentialed is *expected*, because credentials are a retrospective proxy pressed into proactive service, and the proxy false-negatives precisely on the profiles whose making it cannot span — the person-level analog of §7's vacancy discount, a metric-shaped markdown landing on exactly the hard cases. The corollary has kin it should name: signaling theory prices credentials as deliberately lossy proxies [57], the credential-society literature documents the proxy's social career [58], and Goodhart's law describes proxy corruption under targeting [59] — while Meehl's classic case for statistical over clinical prediction [60] sits in no conflict with any of this, its superiority results living in criterion-ready tasks where the output space is already spanned, which is exactly the condition the frontier lacks. The record this paper carries contains one candidate instance in the raw: the 2023 moderation decline preserved in the correspondence archive, asking after a conventional publication record — the metric caught in the act. Whether it issued a *false* negative is not this paper's to declare; by

the standing conditional, that adjudication belongs to the external record the archive exists to enable, and the corollary claims only the base rate: in any regime measured by proxies weaker than the thing itself, such cases must occur.

9 The dataset: a documented genealogy

9.1 The record and its checkability

The genealogy below rests on artifacts a reader can audit rather than trust, and the artifacts are of four grades, each licensing exactly what it licenses. G_1: an independently and publicly timestamped artifact (an archive's version trail, a dated public deposit); G_2: a hash-attested deposited artifact; G_3: a self-hosted or locally timestamped artifact (file metadata, a journal entry on the author's machine); G_4: retrospective memoir. G_1 and G_2 establish existence by a date, and with it temporal ordering against the world; G_3 establishes ordering within the author's own record, checkable for internal consistency and no more; G_4 establishes what the author later believed. None of the four establishes a causal interpretation of the author's own cognition without further evidence, and the annotations below are graded accordingly. The private leg is a run of dated journal entries and documents, several reproduced verbatim in the long manuscript [1] and excerpted below. The public leg is four prints: the manuscript of January 28, 2023, *Centric Unfolding* (`https://aethus.org/aethic-code/files/Jan2023Aethic.pdf`; "January 2023" on its face, the day and second — 9:51:29 AM — by the file's metadata, the same grading as the May print's day); the original short manuscript of May 2023 ("May 2023" on its face, May 23 by the file's metadata; hosted at the framework's site: the print at `https://aethus.org/aethic-code/files/Original_Aethic.pdf`, its phrase-level annotated edition at `https://aethus.org/aethic-code/history/marked_original.html`); the November 2024 preprint frozen as version 1 in PhilArchive's public version trail (`https://philarchive.org/versions/BENARA-5` — every successive deposit dated by the archive, not the author); and the 2026 version with its companions.

The division of labor among the four is itself a finding. The January manuscript is the record's cleanest citation-free exhibit and its sharpest intermediate: all three rules already assembled and numbered — the first verbatim as it reads ever after ("Ethus" for "Aethus"), the second carried by a since-retired "Anti-Ethus" complement construct *defined* by statistical independence, the third in a parent-directed predecessor form, validity checked up the ancestral line rather than down the child line — alongside the soccer-field experiment in manuscript form, a named compression postulate for collapse, permanence already stated of the past Ethus, and not one citation on the page: no references section, no interpretive literature, nothing. The May 2023 manuscript then contains all three postulates in their lasting shape — the third now with its exact two-generation existential–universal alternation, so that a too-strong/too-weak elimination can be read *across dated texts*: the January candidate deployed in earnest, too weak on my reading and kept afterward as a consequence, a too-strong candidate stated in the journal on May 7 and rejected on the May page as "close to the final realization, but still not quite accurate," and the two-generation form arriving by the May print — on May 8, by my inference from the journal entries of May 7 and 9, neither of which records the form reached — along with permanence stated flatly, multi-observer consistency run through the third postulate, and both species of superposition present extensionally on the double-slit cases, insisted to be distinct, months before either had a name — the interval publicly attested at its midpoint, since the October 2023 presentation below still carries the distinction unnamed, and the author dates the naming to the turn of 2024. What

2024 adds is, above all, *names*. And 2024 carries the record's sharpest single certificate: across the deposited text the phrase "possible world(s)" appears fifty-six times while "counterfactual," "Lewis," "Stalnaker," and "perturbation" appear exactly zero times — counts trivially recomputable by any reader, and exactly the lexical shadow one expects of a modal framework built in isolation from the modal literature, though exposure left uncited would cast the same shadow, so the counts certify the absence of the terms and not the isolation.

The public leg also carries an artifact of a different class, sitting in the gap between the May print and the 2024 preprint: a pair of video presentations of October 19, 2023 — the framework's first spoken exposition, Aethic in the first and Nexic in the second — platform-dated rather than self-hosted, the date as the platform displays it, October 20 in some time zones (`https://youtube.com/playlist?list=PL1NA36JDv_T_1gJnzWhevIafOe0x4Q1yC`). Its evidential contribution is precise. The third postulate is delivered there in its finished two-generation form, five months after the May print and thirteen months before the preprint: an existential child whose every child fails to contradict the parent, unchanged. The second postulate still wears the statistical-independence dress. Both species of superposition are again present, again unnamed, and the distinction is put algebraically — one slit and not the other, either way round, asserted not to equal both slits at once — with the speaker flagging in his own voice that the point took him a while to comprehend, which is what a missing name costs a person trying to convey a distinction he can already use. And the modal lexical shadow extends to speech — the channel priced first: the transcript is auto-generated captioning, which garbles the framework's coined vocabulary throughout while leaving standard English terms intact, so the counts that follow are counts over that channel. Across it, "possible world," "counterfactual," "Lewis," "Stalnaker," and "perturbation" occur zero times, as do "many worlds" and "Copenhagen" — a spoken artifact, unscripted so far as the delivery indicates, showing the same absence the deposited prints show.

The record also keeps its second column honestly: the May 2023 second postulate runs on "statistical independence" rather than blankness (the content fixed, on my reading, the dress since evolved); the first postulate arrives, in May, dressed as a generalization of an assumed QBism — an on-ramp, and a dress the record now *brackets*: the subjective-reality and relative-superposition material stands in the dated August 22, 2022 entries of §9.2, the assembled rules stand in the January 28, 2023 print with not one citation on its pages, and the QBism framing appears only at the May write-up — so whatever the exact date of first contact with the interpretive literature (the author reports the months before that print, QBism and relational quantum mechanics among the reading), the content is dated on both sides of the dress's arrival; the mature framework retires the dress for foundations of its own, the states never doxastic in the finished form; an "Anti-Time" construct appears there and nowhere after (retired); and a Rare-Earth application appears that later versions re-house rather than retire — Nexic-wing content, the wings already distinct in the development itself, whose presence without its own context on the page merely risked reading as overreach. A genealogy that can distinguish re-housing from retiring is doing classification, not confession.

And the dataset's two evidential legs are graded explicitly rather than haloed together: from 2018 forward the record is dated artifacts — documents, journal entries, deposits — several checkable publicly and the rest checkable against the session record once it is deposited; the childhood exhibits of §9.2 are retrospective memoir — memoir in grade, even where compiled and hosted (`https://aethus.org/aethic-code/history/ajax-grade-school-ideas.html`, written down in the author's final year of high school; memoir with a URL is still memoir) — and the theses lean on them for texture, never for certification. A third evidential leg outranks both for the dates it touches: a correspondence archive (`https://aethus.org/aethic-code/history/correspondence.html`)

preserves the work's institutional contact, 2023–2026, as DKIM-signed originals with manifest and checksums, hosted with the prints — anchoring those dates to mail infrastructure controlled by neither the author nor the deposits. Its sharpest item, an arXiv account confirmation of May 8, 2023, does not date the completion itself; it dates an account event, to the very day the genealogy marks as the final push, and a journal entry of the next day, self-dated, ties the event to the author's contemporaneous judgment that the work was ready to publish: "yesterday, right after, (at least I hope), officially cracking the measurement problem, I made an arXiv account."

9.2 The stages, annotated

I annotate each stage with the thesis it instantiates, and I begin much earlier than the framework, because the record does.

Preschool. On a Windows XP machine I discovered, by moving to the root directory and descending a different pathway, that the file explorer reaches the control panel — a location I had internalized as a distinct, closed-off domain of the start panel. The thought in the moment: two categorically distinct domains can be welded together beneath the surface, two vantages on a single underlying domain whose distinction lives in your priors rather than in the structure. The inner knowing that arrived with it — *the barrier between any two things is an illusion* — is my first epiphany on record, and the reader will recognize it as the substance/reasoning axis in embryo: the profundity was a deleted boundary, not an added thing.

Kindergarten. I imagined addition as preparing and projecting a number forward on the number line — $2+3$ preparing to move forward from 2 by 3, the equals sign running the command — and then dreamt up an alternative in which one changes not the operation but the *running command*: the same preparation, a different mathematical object to run it, keeping the degrees of freedom intact while localizing the change to the qualitative unfolding of the run — ideally projecting the result off the number line altogether, to the alphabet, or, given enough iterations, to something metaphysical. I had not yet learned subtraction existed; the record thus carries, at age five, the pattern that governs everything after — alien methodological conclusions arrived at simply through lack of exposure to standard practice — and an explicit prelude to the function and set-extension intuitions, arguably to the functor, since the synesthetic emphasis sat on the carried structure. (Two later consequences worth logging: teaching myself dimensional analysis in fifth grade by studying patterns in physics equations solidified the operations-as-structurally-intrinsic view; and abstract algebra in college then felt, in places, like *losing* abstraction — global group operations reading as ad hoc against operations intrinsic to the elements.)

Grade school through high school. A fourth-grade shower thought: wanting 17^2 from a memorized $16^2 = 256$ with no paper, I reasoned that adding sixteen makes sixteen sixteens into seventeen sixteens, and since seventeen sixteens is sixteen seventeens, adding seventeen makes seventeen seventeens — $256 + 16 + 17 = 289$ — the identity $a^2 - (a-1)^2 = a + (a-1)$ found as mental arithmetic. A seventh-grade night intuition, written down the next morning: a formula moving three numbers down a conveyor belt so each takes the next one's place — normalize the first two to zero and one, match the rescaled gap to the next difference, translate — which is linear-and-exponential interpolation generalized, invented without having heard of interpolation. Between a few hundred and perhaps a thousand such identities and consequences accumulated, some fundamental, some not; the fundamental theorem of calculus, for the record's second column, was never among them.

2018–2021: the Nexic wing. A Google Doc of July 24, 2018, 7:55 PM, records the observation that both the origination and the extinction of the dinosaurs are requisites for human existence —

charming-stage natural history, at fourteen, with insights arriving on six-month timescales. By the definitive entry of June 9, 2020, the Accordance Principle stands in writing, and the Rare Earth hypothesis had been intuited independently before I heard its name. Nexic reasoning — built at sixteen and seventeen, its foundation since deposited [34] — is thus the record's first completed structure, and the perfect storm that trained the mind the next wing required.

August 2022: the extrusion week. First contact, August 10 by the author's YouTube watch history, looked up two years downstream (the month attested in the 2024 deposit; export pending): the introductory double-slit video, and during it the speedups-permitted inference of §2. Within the week, the seed of Aethic reasoning arrived in a single eureka moment — synesthetically overwhelming, a new landscape of ontology contained whole in one thought — and its methodological content was the crystallization of *what a superposition is, ontologically*: a structural element of a block universe as designated by ignorance, not a jumble of quantum effects. That crystallization is what licensed the next move safely. Within the week, on August 22 at 1:23 PM, I worked the soccer-field thought experiment in real time in my journal; the entry is reproduced whole in the long manuscript, and its load-bearing stretch, from the 1:42 PM continuation, reads:

> *The Anti-Ethus is simple: it is the set of all possible things which are Rebetary, (Relatively Arbitrary / statistically independent), to the Ethus! [...] if a future iteration of my Ethus can potentially contain a current element of my Anti-Ethus, which at this point is in superposition, then it can't be in superposition at all, because I will eventually come to know it. [...] because some of these paths are more probable than others, the actual probability density function is very much in my Ethus, and this itself is the observed quantum wave function. Cool! Now I just have to figure out how the heck future Ajax's Ethus communicates with my Ethus right now, or if such a communication is just an illusion.*

Annotations, against the finished framework: the entry contains the second postulate in its statistical-independence dress; the future-accessibility argument that seeds the third postulate's temporal reach; and the probability-density observation that becomes the weighting story — while lacking, on its own honest face, the recursive two-generation deduction and the blankness primitive, both later. The two-stage model is here *on the record*: the charming vantage ("future Ajax's Ethus communicates with my Ethus") and the structural resolution months away, unforecastable to its own author.

January–May 2023. January 10: the Severed-Ethus — the strongest Anti-Ethus attributes, physically forbidden from ever joining the Ethus — used to state the extrusion principle's cross-paired cases, the first instance of the instinct that becomes disagreeing superpositions. January 28, 9:51:29 AM by the file's metadata: *Centric Unfolding*, the three rules assembled and numbered for the first time in manuscript — the very three sentences the February entry will box as the old laws — with the soccer-field experiment written up and not one citation on the page. February 8, 4:42 PM, the laws rephrased, journal boxes verbatim: the old laws ("Anything may only be real relative to an Ethus"; "Any attribute of an Anti-Ethus will be in superposition relative to its Ethus"; "An Ethus is invalid if a contradiction is triggered anywhere in the superposition of its parent Ethuses") redesigned into the new (reality relative to an Ethus; superposition iff statistical independence, conditioned on validity; *any child of an invalid Ethus will also be invalid*) — the redesign argued, in the entry itself, on mutual exclusivity of the laws' contents, with the contrapositive of the new second law noticed and prized. May 8: one final push from an overheated freshman dorm room — the phenomenology of sustained digging, a stillness, the structure plucked from the lakebed — and the May print follows, its file produced May 23.

2024–2026. The November 2024 print adds the names — agreeing and disagreeing superposition as titled kinds, the union and extrusion principles — under the lexical shadow certified above; 2026 adds the formalization, the companions, and the collaboration whose own provenance apparatus is §11's subject.

9.3 What the annotations show

Three theses run live. The two-stage model is instantiated *within* the record repeatedly, each stage's author visibly surprised by the next — at nineteen-minute resolution inside the August entry itself, whose 1:23 PM section poses the experiment from the charming vantage and whose 1:42 PM continuation opens "Aw crap. I just thought a little bit more about this thought experiment" (the author surprised by his own next thought, in writing, in real time); the 2022 entry cannot see January 2023; the January print cannot see the February redesign eleven days ahead; February 2023 cannot see the two-generation form; the May print cannot see its own modal face, and the 2024 lexical shadow is consistent with its having been built blind. The substance/reasoning axis runs from the preschool epiphany forward: every recorded advance is a deleted boundary or a relocated constraint, never an added primitive — and the one apparent exception, the early Rare-Earth branch, resolves as re-housed wing content, not substance. And the shades-of-similarity thesis explains the record's own reception surface: a framework that reads as eternalism to presentists and presentism to eternalists, whose 2024 print wore the vocabulary of possible worlds while containing a perturbation semantics no reader could name — the semantics since extracted and deposited as its own paper [31]. The through-line, finally, is the one my father drew in chalk: handed a bare driveway, the three-year-old sat overwhelmed until a winding path was drawn, and then rode for hours — and the reason the path is preferable over the driveway is the same reason the Aethus is preferable over a set, the Accordance Principle over the Drake equation, and the derivative structure of Newton's second law over polynomial interpolation of measured events: *it is about something, and that something is structured.*

10 Who can see past entrenchment: a testable leg

Everything above treats entrenchment as the enemy and taste as the weapon; this short section converts that talk into a hypothesis. The question: is the classical, objectivist-deterministic mindset innate to humans, or taught in infancy by a society that runs on it? The question is factorable. Infant core-knowledge results — object permanence, contact causality — already lean partly innate [61], so the design below should be read as discriminating *beyond* the core-knowledge layer: what it factors is not whether structure is innate but which *metaphysical reading* of that structure is, if any. Take the tree model of §6 seriously and suppose that intake of societal knowledge is mediated by the very capacities the dominant branch is strongest in; then populations differing in that mediation constitute natural comparison groups, and four suggest themselves for maximal differentiation: adults of the dominant cognitive branch in a Western society; adults of other branches, in their subcategories; infant children younger than two; and people of indigenous or otherwise non-Western societies. Two variables separate under such a comparison — a sample's mindset, and the innate human mindset — and the design is stated here as a proposal, not a result.

One reform of my own earlier conjecture belongs on the record, because the orangutan argument of §6 forced it. I had earlier supposed that a highly developed theory of mind might conflict with a generative intuition for the framework's way of thinking; but if the theory-of-mind construct [62]

dissolves under localize-the-flex — if what apes lack is not a faculty but a predictable contextual environment — then the conflict must be relocated accordingly: it is very directly the *cultural implications of language itself*, the standing lexicon of §6, that stands in tension with attaining the framework, not any faculty of mind-reading. On that reform, my own case reads as something of an optimization event — a pre-linguistic mode of perception together with the articulation powers the dominant branch's language rewards, two modes usually in conflict[11] — and the hypothesis above is how the reading stops being about me: if the classical mindset is substantially socialized, then the population of minds positioned to see past it is a measurable feature of the world, and the prescription of §12 to attack the α-space with artists by trade acquires a demography.

11 The modern layer: provenance in the AI era

The genealogy method acquires a new wing in the present decade, and this corpus has run the experiment on itself. The framework's 2026 formalization proceeded within an AI-assisted drafting collaboration, and the apparatus built around it is the justification-side discipline this paper's split firewall demands: dated session records; conversation exports, to be deposited with their SHA-256 hashes, which will fix those bytes from the date of deposit; a pass ledger recording every edit with its verification status and its honest failures; adversarial same-family review used strictly as pre-submission audit — whose own logs document errors originating in every party in the loop[12],

[11] Usually, because otherwise someone would have discovered and propagated the framework ages ago — and I remain genuinely hesitant to claim priority over one population in particular on exactly these grounds: deep thinkers comparatively immune to the dominant branch's entrenchment may have arrived here before me and simply never translated the result across the lexical boundary this paper describes. I state that as the open possibility it is, not as a finding.

[12] One instance is precise enough to belong on the record, and it occurred inside this paper's own review, in the exchange that produced the proximity corollary and its dynamic form (§7); because the instance is itself linguistic masking, the misparse is reconstructed here at the level of the words rather than deferred to the archive. The remark used as a prompt to Claude, verbatim: "I think all of my novelties still count here, because the fact that they're similar but different in key ways to the existing stuff is exactly the shades of similarity in practice," followed by "if I happen to be right about my stuff, then the existing versions being wrong in subtle but unparseable ways is a crisis for them, no?" The reviewer's extraction keyed on the surface chain *count . . . because . . . similar*: it parsed "count" in the evidential register, took the "because"-clause to be offering resemblance as *support*, filed the whole under the nearest established slot the lexicon supplies for that surface — the unconditional fallacy "my work resembles deep work's predicament, therefore my work is deep," the conventionalist stratagem this very section condemns — and read the conditional as governing only the downstream "crisis" consequence. The real meaning was pressured elsewhere in the phrasing: into the antecedent "if I happen to be right," which in the author's intent governed the *count* claim as well; into "similar but different in key ways," which compressed an argument about difficulty terrain — proximity is what obstructed the standing positions from closing the gap, so given correctness the smallness of the deltas measures the terrain's crowding rather than the contribution's size; and into "count" itself, which was doing evaluative work (do not down-weight by resemblance) rather than evidential work (resemblance supports me). The correction named the delta in so many words: "What I said was that IF I AM RIGHT," (capitals his) "then the being close to me in the standard stuff obstructed it from perfectly closing the gap, since I had to hit the whole thing independently" — with the consequence scoped to neighbors "wrong in ways that decisively count for a subset of applications." The classification snapped to correct on the naming, and the dynamic form arrived in the following turn. The mechanics are exactly this section's: the standing lexicon holds a high-prior slot for the surface pattern and no slot at all for the intended content, so the nearest-slot parse won until the delta was stated — misfiling by resemblance under a dense neighborhood, correction by named delta, the exit condition operating as designed, and the lock briefly bidirectional in the manner of the Grothendieck exhibit: the author heard his semantics in his sentence, the reviewer read the sentence under the standing semantics. The full exchange is preserved verbatim in the session record, dated and to be deposited, as corroboration rather than substitute. It is reported as the corollary's first data point rather than as a flourish: the reception dynamics ran as predicted on the thesis itself, and ran in an artificial interlocutor — consistent with masking being a property of inference under density rather than of human psychology in particular, with the caveat kept attached that such an interlocutor is plausibly *more* prone

tallied by round in the provenance record — two same-family branches of the review have, notably, produced divergent readings of which party led the late-round errors, a divergence the dated ledger exists to arbitrate — and thereby document precisely why such review cannot self-certify; and an independent human gate named in every paper as the actual final round. The point for philosophy of science is structural rather than promotional. The art/artwork split of §8 is embodied here as a division of shifts: invention stayed scrappy — rulings made on instinct, structures slid like lenses over prior ontology, the outlaw's shift — while verification was exacting, count-guarded, and logged, the sheriff's. And the provenance record extends the memoir-as-data method into an era that badly needs it: where authorship is about to become chronically contestable, a discipline of dated, hash-attested, publicly deposited records converts the question "who contributed what to the text" from a matter of testimony into a matter of lookup, though not the question where the ideas originated, on which such records are evidence rather than verdict. The session report is already publicly archived with snapshots; the full record, with its hashes, is planned for the same deposit alongside the corpus — and nothing in this paper asks to be believed about the apparatus that cannot be checked against those artifacts.

12 What this buys

Prescriptions, stated as prescriptions — tips of the Vonnegut species under §8's line that heuristics are not axioms, not rules of the making. *Keep genealogies.* The marginal cost of dated records is trivial — a timestamp on a note — and the discovery-theoretic value compounds; this paper's central limitation is that its dataset has one subject, and that limitation dissolves exactly as fast as other builders keep their own records. *Fund disciplinary spontaneity.* If the α-space of accumulated bias is where breakthroughs live, then it can only be attacked by practitioners who know the work will be scrappy — artists by trade — and knowing what to expect ahead of time is exactly the luxury whose absence defines the terrain; institutions that demand forecastable methods are selecting against the only methods that work there. *Inoculate against camouflage.* The shades-of-similarity thesis predicts confident, articulate, mutually inconsistent misreadings of anything deep; a framework can forecast its own predictable garblings in advance and correct them on the page before they occur, and this corpus has practiced exactly that. *Apply the criteria as filters.* The Strong Implication Criterion sorts interpretive proposals now — whatever merely consists with the phenomena, without implying them against alternatives, is incomplete on its face — and the Strong Rooting Criterion, run in contrapositive, points where the next theory must live. *Localize the flex before diagnosing deficits.* Whether the other mind is an ape, a member of another cognitive branch, or a machine, import the full interior and solve for the environment before concluding that a faculty is missing. *Price contributions with vacancy reports.* When deltas read as small against a dense neighborhood, ask how long the spot sat empty and whether anyone ever stated it — an unstated vacancy is camouflage's signature, a stated-but-unengaged one masking's, and a stated-and-rejected one evidence the other way where the rejection engaged the version later accepted — and let the report inform judgment without hardening into a scoring rule, per the art thesis: the column is evidence, not algorithm. And last, the division of labor that the whole essay defends: no rules for the making, exacting rules for the checking — the outlaw and the sheriff, different shifts, one job.

---

to nearest-neighbor misfiling than a careful human referee, so the instance extends the thesis's scope rather than establishing it, and it is, like any first data point, exactly one instance.

A note on provenance. Dates first, since they carry the facts: the criteria, theses, genealogy items, and journal entries collected here date as marked in the text — from 2018 (and, for the childhood exhibits, earlier) through 2026 — with the entries reproduced or excerpted verbatim from the long manuscript [1], where the full versions and that manuscript's own continuity apparatus reside — an apparatus of three prints rather than this record's four, because it predates the January print's surfacing, which occurred only in this record's 2026 assembly (§9.1). This essay collects and develops that material within an AI-assisted drafting collaboration (2026), proceeding from the author's framework, prior manuscripts, and instructions; the first-person voice is the author's by the genre declaration of §1, with the external-case argumentation and the connective prose drafted within the collaboration and author-approved. Every recollection in the genealogy section is author-originated content, at most with drafted wording, checkable against the session record once it is deposited. AI-generated review was used only as pre-submission audit; independent human refereeing precedes any submission; and a dated audit of draft provenance travels in the companion session record.

References

[1] Ajax Benander. *Aethic Reasoning: A Comprehensive Solution to the Quantum Measurement Problem*. Manuscript. Manuscript; the complete development of the framework, successive deposits November 2024–2026. Nov. 2024. Alternative URL: `https://philsci-archive.pitt.edu/id/eprint/25713`. URL: `https://philpapers.org/rec/BENARA-5`.

[2] Ajax Benander. *Auditable Discovery Records: Telling Development from Reconstruction in a Dated First-Person Archive*. Companion paper. Oct. 2026.

[3] Howard E. Gruber. *Darwin on Man: A Psychological Study of Scientific Creativity*. E.P. Dutton, 1974.

[4] Ryan D. Tweney. "Faraday's Discovery of Induction: A Cognitive Approach". In: *Faraday Rediscovered*. Macmillan, 1985.

[5] Frederic Lawrence Holmes. *Hans Krebs: The Formation of a Scientific Life*. Oxford University Press, 1991.

[6] Nancy J. Nersessian. *Creating Scientific Concepts*. MIT Press, 2008.

[7] Thomas (ed.) Nickles. *Scientific Discovery, Logic, and Rationality*. Reidel, 1980.

[8] Norwood Russell Hanson. *Patterns of Discovery*. Cambridge University Press, 1958.

[9] Kevin Dunbar. "How Scientists Really Reason: Scientific Reasoning in Real-World Laboratories". In: *The Nature of Insight*. Ed. by Robert J. Sternberg and Janet E. Davidson. Cambridge, MA: MIT Press, 1995, pp. 365–395.

[10] Kevin Dunbar. "How Scientists Think: On-Line Creativity and Conceptual Change in Science". In: *Creative Thought: An Investigation of Conceptual Structures and Processes*. Ed. by Thomas B. Ward, Steven M. Smith, and Jyotsna Vaid. Washington, DC: American Psychological Association, 1997, pp. 461–493.

[11] Timothy Gowers and Michael Nielsen. "Massively Collaborative Mathematics". In: *Nature* 461.7266 (2009), pp. 879–881.

[12] D. H. J. Polymath. "A New Proof of the Density Hales–Jewett Theorem". In: *Annals of Mathematics* 175.3 (2012), pp. 1283–1327.

[13] Hans Reichenbach. *Experience and Prediction*. University of Chicago Press, 1938.

[14] Karl R. Popper. *The Logic of Scientific Discovery*. Hutchinson, 1959.

[15] Henri Poincaré. *Science and Method*. Thomas Nelson, 1908.

[16] Jacques Hadamard. *An Essay on the Psychology of Invention in the Mathematical Field*. Princeton University Press, 1945.

[17] Michael Polanyi. *Personal Knowledge: Towards a Post-Critical Philosophy*. University of Chicago Press, 1958.

[18] David Bloor. *Knowledge and Social Imagery*. The Strong Programme's reflexivity tenet. Routledge & Kegan Paul, 1976.

[19] Malcolm Ashmore. *The Reflexive Thesis: Wrighting Sociology of Scientific Knowledge*. University of Chicago Press, 1989.

[20] John D. Norton. "General Covariance and the Foundations of General Relativity: Eight Decades of Dispute". In: *Reports on Progress in Physics* 56 (1993), pp. 791–858.

[21] David Deutsch. *The Beginning of Infinity*. Allen Lane, 2011.

[22] Pierre Duhem. *The Aim and Structure of Physical Theory*. Princeton University Press, 1954.

[23] Alberto Montina. "Exponential Complexity and Ontological Theories of Quantum Mechanics". In: *Physical Review A* 77, 022104 (2008). arXiv: `0711.4770`.

[24] Robert W. Spekkens. "Evidence for the epistemic view of quantum states: A toy theory". In: *Physical Review A* 75 (2007), p. 032110.

[25] Robert W. Spekkens. *What is Reality Made Of? Unscrambling the Quantum Omelet*. Interview by Shane Farnsworth, The Escaped Sapiens Podcast, Episode 74. Video available at `https://youtu.be/J2ZIRkfrFlI`. Nov. 2024. URL: `https://youtu.be/J2ZIRkfrFlI`.

[26] E. T. Jaynes. "Probability in Quantum Theory". In: (1990). In Complexity, Entropy, and the Physics of Information (W. H. Zurek, ed.), Addison-Wesley; the omelette passage.

[27] Albert Einstein. "What is the Theory of Relativity?" In: *The Times (London)* (Nov. 1919). The principle-theories versus constructive-theories distinction.

[28] Imre Lakatos. *The Methodology of Scientific Research Programmes*. Cambridge University Press, 1978.

[29] David Schmid, John H. Selby, and Robert W. Spekkens. "Unscrambling the omelette of causation and inference: The framework of causal-inferential theories". In: *arXiv:2009.03297* (2020).

[30] Thomas S. Kuhn. *The Structure of Scientific Revolutions*. University of Chicago Press, 1962.

[31] Ajax Benander. *Counterfactual Evaluation by Perturbation of Information States: Deleting the World-Similarity Ordering, Not Refining It*. Preprint. 2026. URL: `https://philpapers.org/rec/BENCEB-2`.

[32] Ajax Benander. *Quantum Mechanics from Records: Three Consistency Postulates, One Composition Primitive, and One Kernel Recover the Measurement Architecture of the Dirac–von Neumann Formalism, with the Projection Postulate as a Theorem*. Preprint. The physics-face axiomatization; first posted July 17, 2026. 2026. URL: `https://philpapers.org/rec/BENTQB`.

[33] Ian Hacking. *The Social Construction of What?* Harvard University Press, 1999.

[34] Ajax Benander. *The Copernican Method Against the Copernican Principle: Anthropic Typicality over Partial Information States*. Preprint. 2026. URL: https://philpapers.org/rec/BENTCM-7.

[35] David J. Chalmers. "Verbal Disputes". In: *Philosophical Review* 120.4 (2011), pp. 515–566.

[36] Richard P. Feynman. "There's Plenty of Room at the Bottom". In: *Engineering and Science* 23.5 (1960), pp. 22–36.

[37] Allyn Jackson. "Comme Appelé du Néant — As If Summoned from the Void: The Life of Alexandre Grothendieck". In: *Notices of the AMS* 51.9–10 (2004).

[38] Alexander Grothendieck. *Récoltes et Semailles*. Circulated 1986; published Gallimard, 2022. 1986.

[39] Joseph Jordania. *Who Asked the First Question?* Logos, 2006.

[40] David Premack and Ann James Premack. *The Mind of an Ape*. W. W. Norton, 1983.

[41] Paul Hoyningen-Huene. "Context of Discovery and Context of Justification". In: *Studies in History and Philosophy of Science* 18.4 (1987), pp. 501–515.

[42] Paul Hoyningen-Huene. "Context of Discovery versus Context of Justification and Thomas Kuhn". In: *Revisiting Discovery and Justification: Historical and Philosophical Perspectives on the Context Distinction*. Ed. by Jutta Schickore and Friedrich Steinle. Archimedes 14. Dordrecht: Springer, 2006, pp. 119–131.

[43] P. Kyle Stanford. *Exceeding Our Grasp: Science, History, and the Problem of Unconceived Alternatives*. New York: Oxford University Press, 2006.

[44] P. Kyle Stanford. "Darwin's Pangenesis and the Problem of Unconceived Alternatives". In: *The British Journal for the Philosophy of Science* 57 (2006), pp. 121–144.

[45] P. Kyle Stanford. "Francis Galton's Stirp Theory of Inheritance and the Problem of Unconceived Alternatives". In: *Biology and Philosophy* 21 (2006), pp. 523–536.

[46] P. Kyle Stanford. "August Weismann's Theory of the Germ-Plasm and the Problem of Unconceived Alternatives". In: *History and Philosophy of the Life Sciences* 27 (2005), pp. 163–199.

[47] Abraham Pais. *Subtle is the Lord: The Science and the Life of Albert Einstein*. Oxford: Oxford University Press, 1982.

[48] Gerald Holton. *Thematic Origins of Scientific Thought: Kepler to Einstein*. Revised. Cambridge, MA: Harvard University Press, 1988.

[49] Ajax Benander. *Sequential Achievement and the Sharp Crossover in Outlier Productivity*. Preprint. 2026. URL: https://philpapers.org/rec/BENSAA-11.

[50] "Presume It Not: True Causes in the Search for the Basis of Heredity". In: *The British Journal for the Philosophy of Science* 71.1 (2020), pp. 59–86.

[51] Kurt Vonnegut. *Bagombo Snuff Box*. G.P. Putnam's Sons, 1999.

[52] Herbert A. Simon. "Does Scientific Discovery Have a Logic?" In: *Philosophy of Science* 40.4 (1973), pp. 471–480.

[53] Pat Langley et al. *Scientific Discovery: Computational Explorations of the Creative Processes*. MIT Press, 1987.

[54] Carl G. Hempel. “Studies in the Logic of Confirmation”. In: *Mind* 54.213–214 (1945), pp. 1–26, 97–121.

[55] Nelson Goodman. *Fact, Fiction, and Forecast*. Harvard University Press, 1955.

[56] Sabine Hossenfelder. *Lost in Math: How Beauty Leads Physics Astray*. Basic Books, 2018.

[57] Michael Spence. “Job Market Signaling”. In: *The Quarterly Journal of Economics* 87.3 (1973), pp. 355–374.

[58] Randall Collins. *The Credential Society: An Historical Sociology of Education and Stratification*. New York: Academic Press, 1979.

[59] Charles A. E. Goodhart. “Problems of Monetary Management: The UK Experience”. In: *Papers in Monetary Economics*. Vol. I. Reserve Bank of Australia, 1975.

[60] Paul E. Meehl. *Clinical versus Statistical Prediction: A Theoretical Analysis and a Review of the Evidence*. Minneapolis: University of Minnesota Press, 1954.

[61] Elizabeth S. Spelke and Katherine D. Kinzler. “Core Knowledge”. In: *Developmental Science* 10.1 (2007), pp. 89–96.

[62] David Premack and Guy Woodruff. “Does the Chimpanzee Have a Theory of Mind?” In: *Behavioral and Brain Sciences* 1.4 (1978), pp. 515–526.